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TECHNICAL COMMUNICATION

A guide to Open-JIP, a low-cost open-source chlorophyll fluorometer

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Abstract
Chlorophyll *a* fluorescence is the most widely used method to study photosynthesis and plant stress. While several commercial fluorometers are available, there is a need for a low-cost and highly customisable chlorophyll fluorometer. Such a device would aid in performing high-throughput assessment of photosynthesis, as these instruments can be mass-produced. Novel investigations into photosynthesis can also be performed as a result of the user's ability to modify the devices functionality for their specific needs. Motivated by this, we present an open-source chlorophyll fluorometer based on the Kautsky induction curve (OJIP). The instrument consists of low-cost, easy-to-acquire electrical components and an open-source microcontroller (Arduino Mega) whose performance is equivalent to that of commercial instruments. Two 3D printable Open-JIP configurations are presented, one for higher plants and the other for microalgae cells in suspension. Directions for its construction are presented and the instrument is benchmarked against widely used commercial chlorophyll fluorometers.

Keywords Chlorophyll *a* fluorescence · OJIP · *Chlorella vulgaris* · Cyanobacteria · Fluorometer · Open-source

Abbreviations

F_V/F_M	Maximum quantum yield of photosystem II
DAQ	Data acquisition device
PAM	Pulse amplitude modulated
Rf	Feedback resistor

Introduction

Chlorophyll *a* is the primary photosynthetic pigment found in all oxygenic photosynthetic organisms (Blankenship 2002). In phototrophic organisms, part of the energy that is absorbed by chlorophyll molecules is converted into biochemical energy and used to drive photosynthesis (photochemistry). However, a portion of the absorbed energy is either dissipated as heat (non-photochemical quenching) or re-emitted as chlorophyll *a* fluorescence. Through the assessment of chlorophyll *a* fluorescence signal, inferences can be made regarding the functioning of photosynthetic pathways (Maxwell and Johnson 2000). These inferences provide an estimate of the health of the phototrophic organism (Lichtenthaler 1988).

The foundation of chlorophyll *a* fluorescence and its interpretation is commonly based on the work of Duysens and Sweers (1963). They proposed that the intensity of chlorophyll *a* fluorescence corresponds to the redox state of the functional photosynthetic units (referring in particular to the reaction centre of Photosystem II). For example, when all PSII reaction centres are oxidised (such as when the plant is in the dark), the fluorescence yield would be at a minimum. However, once exposed to light, PSII reaction centres would become reduced resulting in higher chlorophyll *a* fluorescence. Based on this phenomenon, plant scientists have proposed several parameters to quantify the plants photophysiological state. The most widely used parameter being the maximum quantum yield for primary photochemistry, commonly referred as F_V/F_M (Maxwell and Johnson 2000).

Three main fluorometric approaches exist for the assessment of chlorophyll *a* fluorescence: (1) Pulse Amplitude Modulated (PAM), (2) Fast Repetition Rate (FRR) and (3) Fast rise of the chlorophyll a fluorescence or OJIP

Harvey Bates and Alonso Zavafer have contributed equally to this project.

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fluorometry. Each method integrates different aspects (temporal, spectral and spatial) of the chlorophyll *a* fluorescence signature. The focus of this work will be on OJIP (acronym based on the presence of three main phases O-J, J-I and I-P) fluorometry. OJIP fluorometry uses an excitation light source to drive photosynthesis, which concurrently triggers chlorophyll *a* fluorescence emission. The O-J (photochemical) phase is well characterised and corresponds to a reduction of PSII's primary acceptor (Q_A). The thermal phases (J-I and I-P) are still debated amongst researchers, but generally the J-I phase corresponds to a reduction of the plastoquinone pool (PQ-pool) and I-P refers to a saturation of electrons on the acceptor side of photosystem I (PSI) (Stirbet 2011). In case of some types of stress such as high temperatures, a rapid change in phase can appear in the OJIP transients, this is known as a K-step, corresponding to degradation of the oxygen-evolving-complex (OEC) (Strasser 1997). In order to reduce all PSII reaction centres, the sample must be illuminated with a saturating irradiance higher than 1800 µmol of photons m^{-2} s^{-1} (Strasser et al. 2000; Maxwell and Johnson 2000). The result of this measurement is the rapid polyphasic rise of chlorophyll *a* fluorescence.

Typically, an OJIP fluorometer is composed of a PIN photodiode as a detector and an LED (or LED array) as an excitation source (Strasser 1992). The photodiode is protected by a band-pass (~ 680 nm) or long-pass (> 680 nm) filter which allows only chlorophyll *a* fluorescence to pass. Incoming photons from the filtered chlorophyll *a* fluorescence are detected by the photodiode, which is then measured by a data acquisition device (DAQ) with a typical digitization between 12 and 16 bits. The technical challenges of these instruments are that (1) the detector and DAQ have to be fast enough to detect the fluorescence intensity during the first 100 µs of illumination, this is to assess the value of the fluorescence level when all PSII are oxidised and open (F_O); (2) The photodiode must be sensitive to near infrared light (Maxwell and Johnson 2000).

There is a great diversity of chlorophyll *a* fluorometers on the market; however, the minimum cost of a commercial device is at least 1000 USD (price in June 2019). When multiple units are required, such as with high-throughput phenotyping systems, this becomes an expensive option. In addition, most commercial instruments have a limited degree of customisation both from a hardware and software perspective, which limits the experimental design. For this reason, in this work, we present a fully customisable chlorophyll *a* fluorometer for OJIP and quantum yield assessment, for which construction cost ranges between 50 and 100 USD (Supplementary Table 2). To do so, we have integrated 3D printing and low-cost off-the-shelf electronics to develop a simple open-source OJIP fluorometer (Open-JIP), which can be assembled and used without prior coding or electronics experience. We have designed two types of device, Open-JIP (Algae) for microalgal cells in suspension and Open-JIP (Plant) for plants.

Materials and methods

Setup and construction

Both designs of Open-JIP (Algae/Plant) were modelled in Fusion360 (AutoDesk, California, USA) and 3D printed in black PLA plastic (inkStation, Sydney, AUS; Fig. 1; models are available to download at https://github.com/Open-JIP/Open-JIP). Open-JIP (Plant) was designed around a quick-action clamp (Quick-Grip Medium Bar Clamp, Irwin, Box Hill, Victoria) for which the 3D printed models can be mounted, allowing the user to measure OJIP transients without removing or damaging the leaves (Supplementary Fig. 5).

An Arduino microcontroller (Arduino, Mega 2560 Rev3, Somerville, USA) was used to control both Open-JIP configurations (Algae and Plant, see Fig. 1). This microcontroller was chosen because of its 256-kB memory, which is needed to capture the fluorescence readings along with corresponding timestamps. The microcontroller has a worldwide distribution, which makes it easily accessible for anyone interested in photosynthesis.

Open-JIP (Algae) employs a 90° angle between the detector and emitter, which is used by many commercial chlorophyll fluorometers (Fig. 1a). The excitation source consists of a 10 W LED (COB SMD Chip) of various wavelengths: 466 nm, 532 nm, 592 nm, 626 nm (Supplementary Fig. 1) that were filtered through an IR cut-off filter (IR Filter, Lilly Electronics, Q-BAIHE, Hubei Province, ROC) to remove infrared light emitted from the LED. A PIN photodiode (BPW24R, Vishay, Pennsylvania, USA) was used as a detector and is shielded from stray light by a 695 nm long-pass filter (Thorlabs, New Jersey, USA) allowing only chlorophyll *a* fluorescence to pass through. Styrofoam mirrors (Polystyrene Reflective Mirror, Kadmium, Sydney, Australia) are positioned around the outside of the cuvette (111137, Globe Scientific, Mahwah, USA) resulting in increased signal-to-noise (Supplementary Fig. 2) by reflecting both chlorophyll *a* fluorescence and excitation light.

Open-JIP (Plant) uses a 45° angle between the leaf in reference to both the detector and the emitter (Fig. 1b). The excitation and detection system were identical to the algae version with the exception that the excitation source had a wavelength of 466 nm and no infrared cut-off filter was required as the blue LED generated negligible infrared radiation. The aperture can be changed, based on type/shape of leaf being measured by modifying the 3D printed model, as this will expose less or more PSIIs to the excitation irradiance resulting in increased or decreased signal to noise.

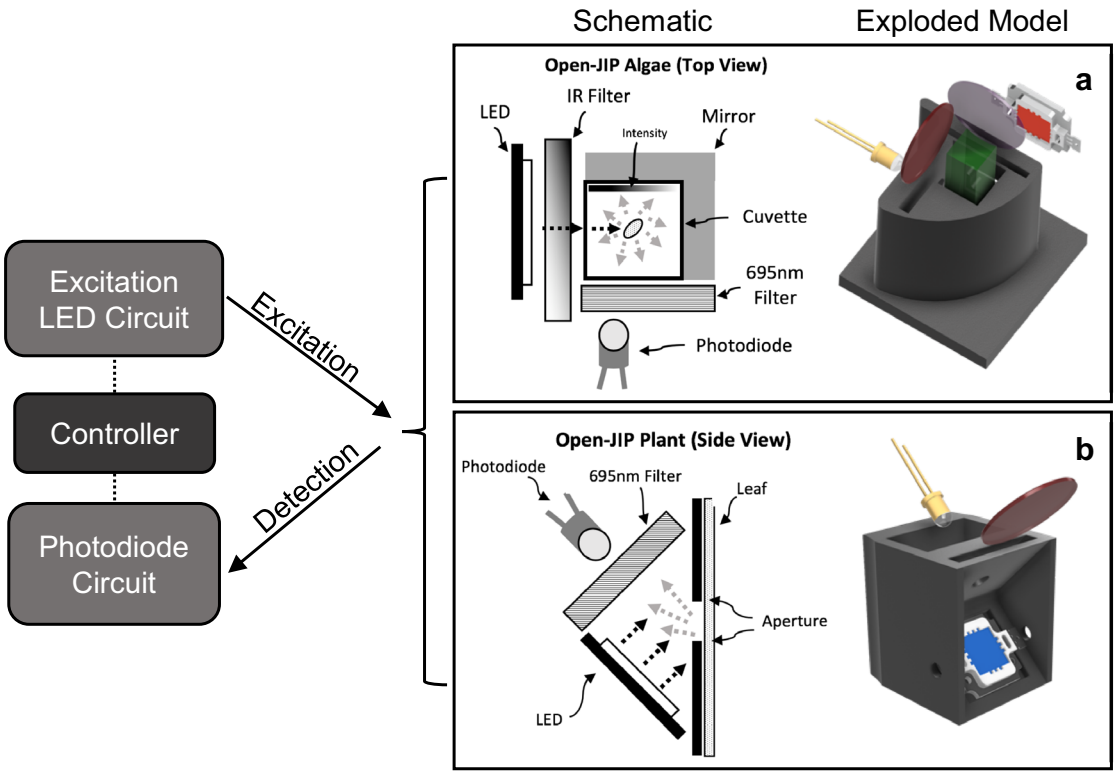


Fig. 1 Block diagram of the Open-JIP unit. The controller sends electrical signal to excitation LED circuit resulting in chlorophyll excitation. The photodiode circuit detects and amplifies the fluorescence signal and sends this information to the DAQ (controller). Both **a** Open-JIP (Algae) and **b** Open-JIP (Plant) are shown; however, excitation for the two devices is operated separately (see https://github.com/Open-JIP/Open-JIP for operating information). Exploded models for each of the versions show the components positioning and the 3D printed scaffold

To detect chlorophyll *a* fluorescence, an operational amplifier (LM358, Texas Instruments, Dallas, USA) was chosen due to its ability to use a single power supply (12 V) and its low power consumption. To adjust the sensitivity of the device, the resistor value (R1) can be switched to a higher resistance for more sensitivity (very dilute cultures) or lower resistance for less sensitivity (concentrated cultures). While a variable resistor could be implemented here, the parallel requirement of changing the capacitor (C1) to prevent overshooting of the operational amplifier would negate its customisability. Future versions of the Open-JIP circuit may provide additional gain settings. Additionally, the gain of the device can be adjusted through the software of Open-JIP. Specifically, this can be performed by changing the Arduino internal reference voltage to either 1.1 V, 2.56 V or the default 5 V with a simple one-line command in the Arduino integrated development environment (IDE).

Open-JIP's excitation LED circuit is powered by a 15 V 1.2 A barrel plug switching power supply (15DYS618, Ideal Power, Texas, USA, see Fig. 2b). To obtain constant brightness of the excitation LED, the 15 V coming from the power supply passes through a 12 V linear regulator (KA7812ETU, On Semiconductor, Arizona, USA). Although the switching power supply could provide power to both the LED and photodiode circuits, the sensitivity of the latter requires a more stable DC supply. This resulted in the use of a 3.7 V 2.4 Ah Polymer Lithium Ion (LiPo) battery to power the sensitive components (photodiode circuit) of the fluorometer (Fig. 2a). The voltage of the battery was boosted using a DC-DC boost converter to ~ 18 V (DFR0123, DFRobot, Shanghai, ROC) and then regulated to 12 V using a linear regulator (L7812CV, Jameco, California, USA). This serves two purposes: (1) being that a higher voltage applied to the photodiode circuit (the operational amplifier in particular) provides increased resolution of the chlorophyll *a* fluorescence and (2) as the battery loses charge the gain of the operational amplifier will remain stable at 12 V.

When a fluorescence measurement is initiated by the user, the microcontroller (Arduino Mega) sends a signal to activate a transistor (T1), resulting in the LED switching on. The switching on (rise time) of the LED must be faster than 50 µs to ensure that a F_O value (minimum fluorescence intensity) can be obtained which is not a result of low actinic light (< 1800 µmol photons m^{-2} s^{-1}). To achieve this, the Darlington Transistor (BD681, Farnell, Leeds, UK) (T1) in conjunction with a parallel diode and resistor arrangement

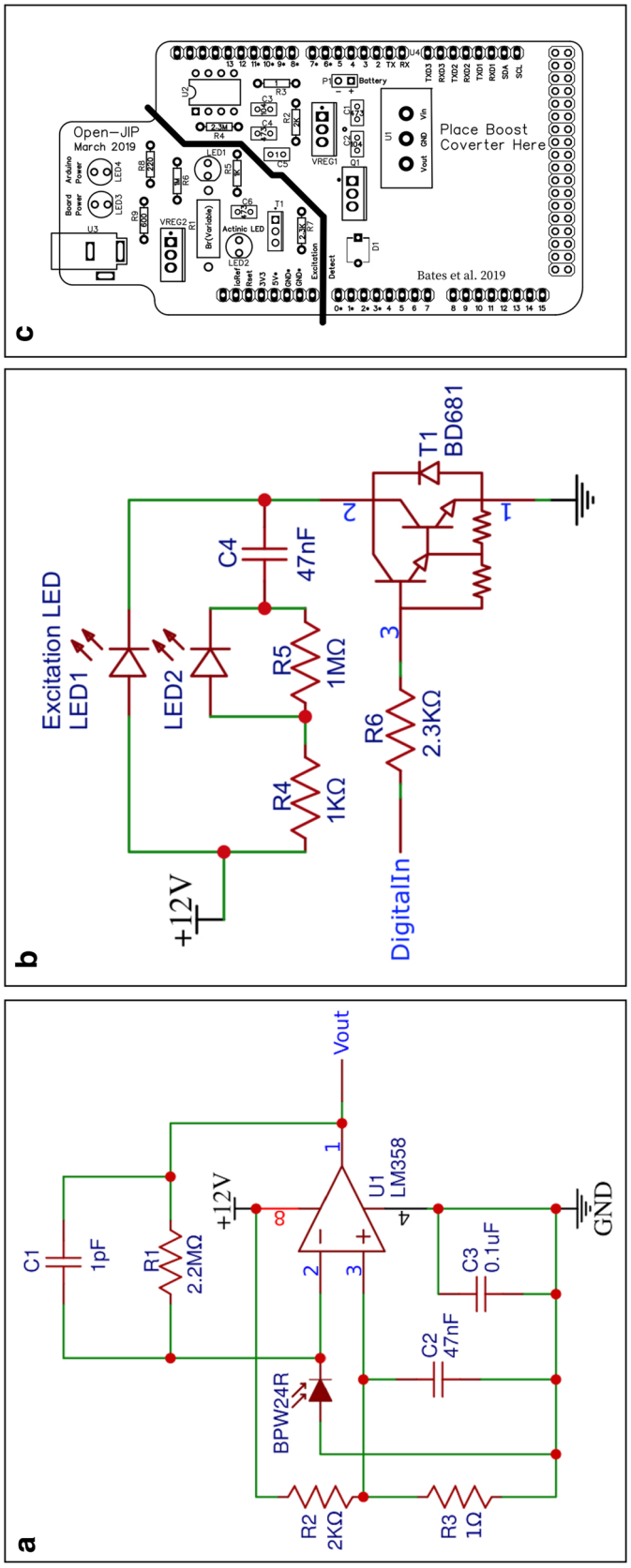


Fig. 2 Circuit diagram of the detector (**a**) and the excitation (**b**) units which are controlled by the Arduino. Schematic (**a**) demonstrates the detector componentry surrounding a LM358 operational amplifier (U1) which is used to amplify the fluorescence signal. Schematic (**b**) shows the componentry used to switch (T1) on the excitation LED (LED1). The excitation LED (LED1) switches on and off the actinic light when the user wants to measure an OJIP curve. Printed circuit board (PCB) design (**c**) allows the user to use the fluorometer without modification (For full schematic see Supplementary Fig. 3)

was used; resulting in a rise time of ~40 µs. The parallel arrangement also prevents a bright flash from the excitation LED when switching on and gives a steady excitation intensity for the entire 1 s measurement. For up-to-date versions of the Open-JIP, circuitry see our GitHub page (https://github.com/Open-JIP/Open-JIP).

Benchmark tests

To demonstrate the device's accuracy, reproducibility and sensitivity, a number of tests were conducted comparing Open-JIP's measurements to a commercial chlorophyll fluorometer (FL3500, PSI, Drasov, Czech Republic).

Biological material

Microalgae of two different taxa were chosen, (1) *Chlorella vulgaris* and (2) *Synechococcus* sp. (CS-29). *C. vulgaris* was grown in MLA medium at 20 µmol photons m^{-2} s^{-1} under white LED tubes, measured with a 2π light meter (LI-250A, LiCor, Nebraska, USA) (Bolch and Blackburn 1996). *Synechococcus* sp. was grown under the same lighting conditions using Pro99 Media (Moore et al. 2007). Three different species of higher plants were chosen (a) *Spathiphyllum wallisii*, (b) *Citrus hystrix* and (c) *Spinacia oleracea* as they present vastly different OJIP transients and demonstrate Open-JIP's ability to quantify these transients.

Experimental design

Aliquots (4 mL) were taken from a stock solution of *C. vulgaris* during its exponential phase and transferred into 10 × 10 mm disposable fluorescence cuvettes. Samples were placed in both the commercial instrument (FL3500) and in the Open-JIP (Algae) which were calibrated to an actinic light intensity of ~2100 µmol photons m^{-2} s^{-1}. These samples were given 5 min to dark-adapt before simultaneously measuring OJIP curves for 1 s in each of the two devices, after which the cuvettes were gently shaken to ensure the algae remained suspended in solution and another 5 min were given to allow the algae to dark-adapt before repeating the experiment two more times. The experiment was then repeated with *Synechococcus* sp.

As a complementary test, changes in chlorophyll *a* fluorescence induction were induced by using a heat treatment with a *C. vulgaris* culture, as demonstrated previously by Tóth et al. (2009) with the higher plant *Arabidopsis*. Six aliquots of *C. vulgaris* were placed inside an oven at 60 °C for 12 min. OJIP kinetics were then measured with the same technique as described above using both instruments ($n = 3$).

Changes in the O-J kinetics were induced by illuminating *C. vulgaris* cells with different levels of saturating light for 5 min, between each of four light intensities. In order to do this, neutral density filters were added to Open-JIP in front of the excitation LED allowing for four different actinic light (626 nm) irradiances (14,000, 8500, 6000 and 3200 µmol photons m^{-2} s^{-1}) as measured by a 4π light meter (LI-250A, LiCor, Nebraska, USA).

Open-JIP (Plant) was used to obtain OJIP transients on three species of plants, *S. wallisii*, *C. hystrix* and *S. oleracea*. A 5-min interval was left between the three replications to allow the samples to dark-adapt. Unlike the tests done on microalgae, this experiment used a blue (466 nm) excitation LED to show the devices ability to use multiple wavelengths to assess OJIP.

Data analysis

Recorded OJIP curves are copied from the serial monitor of the Arduino software (Arduino IDE, Somerville, USA) and formatted into a.CSV file in excel. Relative variable florescence (V_t) was calculated as

$$V_t = \frac{(F_t - F_o)}{(F_m - F_o)} \quad (1)$$

Allowing the comparison of the OJIP kinetics between the two devices, as they have different scales for measuring chlorophyll *a* fluorescence. Data were then plotted using Python (see https://github.com/Open-JIP/Open-JIP for example scripts).

Results

A comparison of the OJIP curves of *Synechococcus* sp. and *C. vulgaris* obtained by Open-JIP and a commercial instrument (FL3500 chlorophyll fluorometer) demonstrates the device's ability to produce accurate and reproducible analysis of the OJIP transient (Fig. 3). As shown in Fig. 3, the transients measured the commercial (FL3500) and the Open-JIP (Algae) throughout comparable for all samples (for quantum yields see Supplementary Table 1). Slight variations can be noted in the O-J rise of Fig. 3f which is the result of different actinic light intensities [as seen by Schansker et al. (2011)] between the two instruments (Fig. 4a). These variations cannot be fully avoided, due to the potential differences in optical configurations between instruments. However, this has no impact on the interpretation of the obtained OJIP curves and data analysis. In an effort to make it as comparable as possible, we have used an LED with similar wavelength and intensity.

To test the time response of Open-JIP, the effect of light intensity on the rise of O-J phase in *C. vulgaris* was measured (Fig. 4a). As the actinic light intensity of the

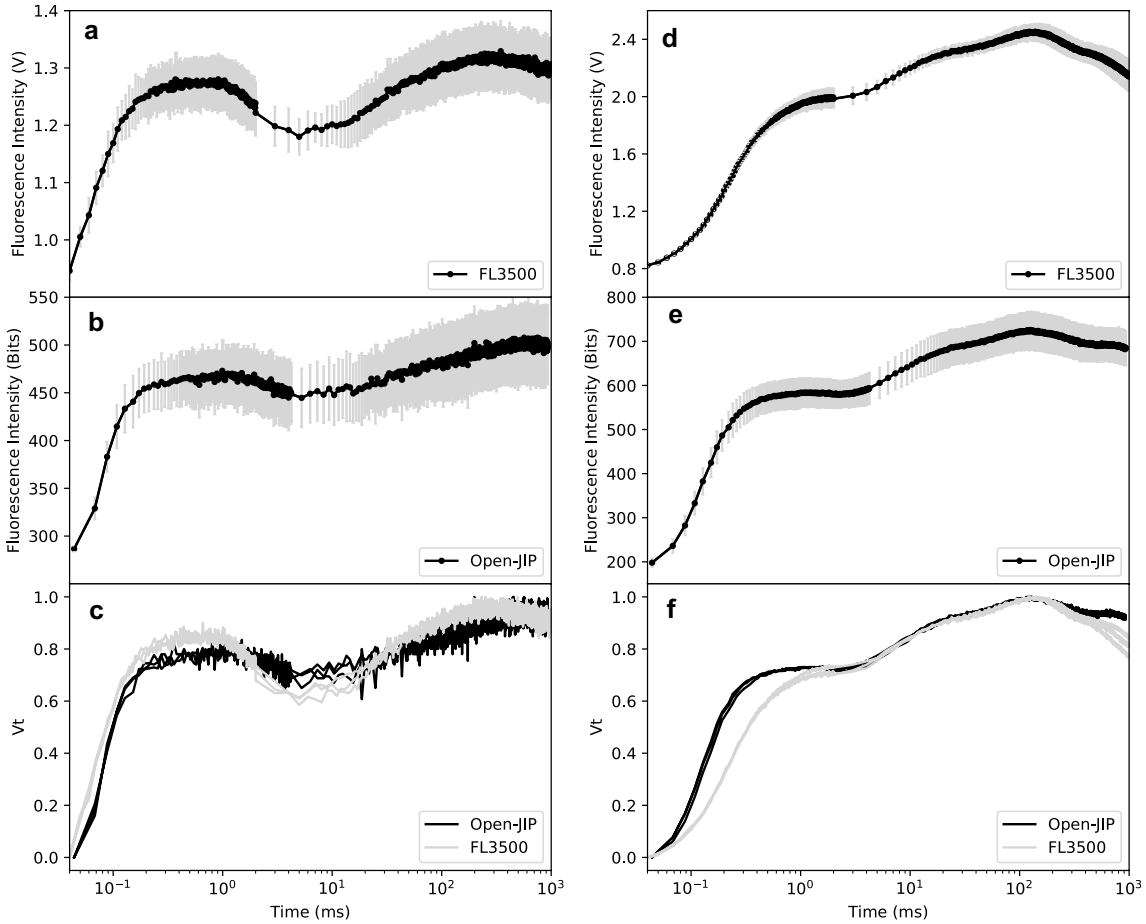


Fig. 3 OJIP transient for *Synechococcus* sp. *(CS-29)* (**a**, **b**, **c**) and *C. vulgaris* (**d**, **e**, **f**). **c** and **f** Demonstrate relative variable fluorescence (V_t) for each of the two microalgae and instruments with three replicates of each. Average ($n = 3$) and standard deviations are shown for **a**, **b**, **d** and **e**

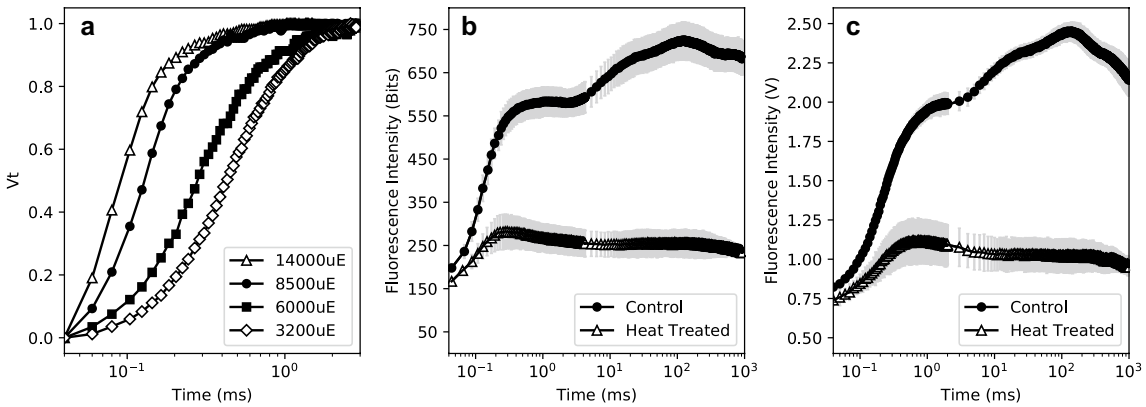


Fig. 4 Effect of different light intensities (**a**) on the O-J rise of chlorophyll a fluorescence in *Chlorella vulgaris* as measured by Open-JIP (Algae). Relative florescence intensity (Vt) is shown and normalised to 3 ms to better demonstrate this feature. **b** Non-heat treated (black circles) and after 12 min at 60 °C (white triangles) of the OJIP measured with Open-JIP Algae with *C. vulgaris* ($n = 3 \pm$SD). **c** Same experiment as in (**b**), however, OJIP transients are measured with the FL3500 fluorometer ($n = 3 \pm$SD)

fluorometer increases, so too does the rate at which fluorescence rises for the O-J phase. This agrees with previous observations by Schansker et al. (2011), indicating that the O-J phase is strongly dependent on functional cross section of PSII. The fact that we observed these changes on the O-J phase indicates that Open-JIP has sufficient time resolution

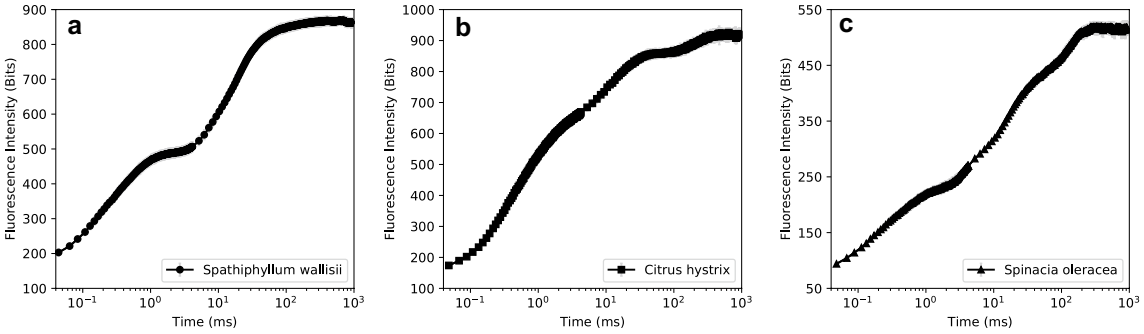


Fig. 5 Open-JIP (Plant) OJIP transients of different species of plants. **a** *Spathiphyllum wallisii* (black circles), **b** *Citrus hystrix* (black squares), **c** *Spinacia oleracea* (black triangles). OJIP transient measured with Open-JIP (Plant) using a 466 nm actinic light. Average ($n = 3$) and standard deviation are shown in grey

to measure the fast components of the chlorophyll fluorescent transients.

To provide evidence that Open-JIP can be used to assess different phenotypes in photosynthetic organisms, a high-temperature stress treatment was implemented. *C. vulgaris* was exposed to a dark incubator at 60 °C for 12 min as shown in Fig. 4b, c. Control *C. vulgaris* cells exhibit a three-phase OJIP curve using both instruments (commercial and Open-JIP). As expected, the heat-treated sample showed alterations in the OJIP kinetic with both instruments. The formation of the so-called K-step around 300 µs was detected accurately in both instruments, with the K-step being associated as a clear photosynthetic phenotype in the event of high temperatures (Nagy et al. 2012).

Different species of common vascular higher plants were chosen to demonstrate Open-JIP (Plant)'s ability to resolve various OJIP transients (Fig. 5). As can be seen, each plant displays vastly different polyphasic rises, each of which can be measured accurately using Open-JIP (See supplementary Fig. 4 for commercial instrument comparison). Note that Open-JIP system is capable of quantifying variations in photosynthesis using only 10 bits of digitization. Commercial instrument in the market has digitisation of at least 12 bits, but Open-JIP shows equivalent results to commercial instruments. This shows the potential of Open-JIP as a research tool.

Discussion

In the present work, we have shown the feasibility of building an OJIP fluorometer for less than 100 USD. Open-JIP has demonstrated accuracy and time resolution that is comparable to commercial chlorophyll fluorometers. A commercial open-source microcontroller (Arduino Mega) is suitable for this work and acts as a microcontroller, DAQ and PC interface. We can still resolve the main features of the OJIP curve and accurately estimate F_v/F_m. It is important to note that other Arduino boards such as UNO, Micro or Nano are not suitable for this task due to limitations of internal memory.

The Open-JIP fluorometer has many advantages when compared to commercial instrumentation. The instrument's low cost, allows for its mass production and can be used in high-throughput experimental designs at a limited cost. In addition, the ability to fully customise the devices software and hardware (due to it being open-source) allows users to adapt it to their own experimental designs (see https://github.com/Open-JIP for example, modifications). Finally, the simplicity of the device allows users with no electronics and/or programming knowledge to construct and use the instrument.

While the aforementioned advantages would make it attractive for some researchers, one has to bear in mind that the Open-JIP is not intending to substitute the use of commercial instruments. The user may keep in mind that a source of variability in any Open-JIP would be cables used in its construction and different brands of components. Therefore, one needs to have at least one commercial instrument to cross calibrate the Open-JIP and to verify its accuracy. Another limitation is that Open-JIP has a reduced digitisation depth (10 bits, 1024 values) in contrast to commercial instruments, which have depths of 12 to 16 bits (4096–65,536 values), meaning that the resolution of the Open-JIP fluorometer is inferior to commercial OJIP fluorometers; however, this is not a serious limitation considering the device's low cost.

While in some laboratories, it is common to build in-house instruments, fully open-source chlorophyll fluorometers are far less common. For example, Hoadley and Warner (2017) have shown that it is feasible to build a low-cost multicolour FRR fluorimeter for less than 700 USD. The higher cost of that instrument when compared to Open-JIP is due to its higher complexity as it uses a more sophisticated DAQ and optics. Another earlier example is the work of Leeuw et al. (2013) who developed a phytoplankton fluorescence device with an estimated cost of 150 USD; however, the intention of this instrument was to study growth.

The development of Open-JIP provides a customisable low-cost analytical tool for biologists. This is the main benefit of the instrument, as novel biophysical applications require novel and modifiable instrumentation to suit the researchers' requirements. In particular, the customisability of the device would allow scientists in the future to add, modify and explore other protocols that will aid understanding the complexities of the OJIP polyphasic rise. Examples of these modifications could include different wavelengths, sampling rates, optical configurations, etc.

The main purpose of Open-JIP is to be used as a low-cost sensor for high-throughput measurements, for example, in algal phenomics whereby the user requires several fluorometers to measure phenotypes simultaneously (Houle et al. 2010). Open-JIP is an example of open-source scientific instrumentation that will enhance the outreach of the OJIP method to a wider group of users in developing countries, citizen scientists and students.

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